

Molecular dynamics simulations of membrane peptide binding and assembly toward rational design

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Rational design of membrane peptides and proteins requires evaluating not only their final structures but also the dynamic pathways through which they fold and assemble. Molecular dynamics (MD) simulations offer a means to evaluate the accessibility of transient states within these processes, which are challenging to characterize experimentally. Here, we examine membrane binding, insertion, and assembly of designed ion-channel peptides and membrane-binding peptides with intrinsically disordered regions. Combined with experimental observations, the simulations reveal how sequence features and membrane properties, including charged residues, membrane potential, and lipid composition, modulate access to different structural states. These insights would provide a basis for integrating dynamic-state evaluation into membrane peptide and protein design.